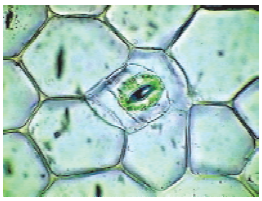
Chapter 6

Fundamental Unit of Life : Cell



We have studied about the diversity in shape and size of cells of both unicellular and multicellular forms of organisms. We also studied that, in the year 1665, Robert Hook gave the name 'cell' to those compartments that he observed in a thin section of cork. Let us study the arrangement and functions of cells with the help of some activities.

6.1 Observing Different Types of Cells

We would require the following things to carry out our observations.

Blade, slide, coverslip, matchsticks, Rheo leaf, ice-cream spoon, red ink, microscope

Activity-1

Observing cells in a section of matchstick.

Take a matchstick and soak it in water for about an hour. Now use a sharp blade to cut a thin transverse section of it. Keep this section on a drop of water, put a coverslip on it and observe it under the microscope. Make a sketch of what you observe. Compare your observation / sketch with figure 1. This figure shows us the sketch that Robert Hooke made in the year 1665, after observing a section of cork under the microscope.

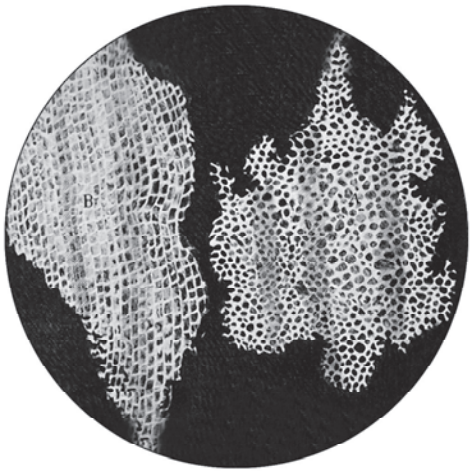


Fig. 1: Section of cork

Activity-2

Observing the outer layer of cells of a leaf

(Activity-2 and 4 may be done together)

Peel off a thin layer from the lower side of the leaf of Rheo. Keep this on a slide, put a few drops of water on it and spread it out evenly. Observe it under the microscope and try to make a sketch of your observation.

Compare it with figure-2 and try to answer the following questions.

- Do all the cells appear same?
- What parts are visible in the cell under the microscope?

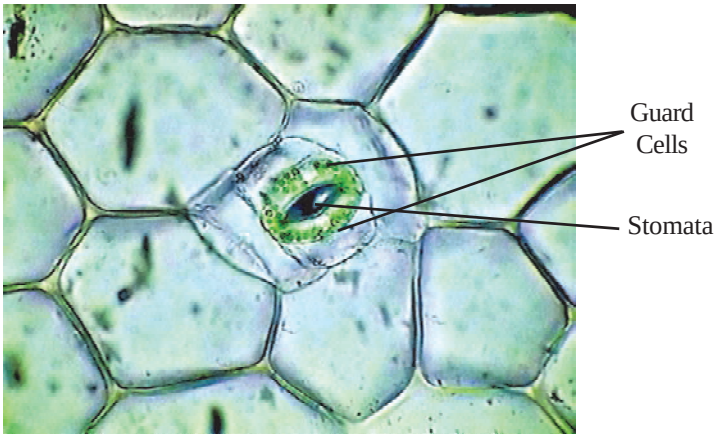


Fig. 2: Cells in the peel of Rheo leaf

You could do this experiment using the leaves of amorphophallous (zimikand) or china rose (Jasone) in case you don't have Rheo (The shape of stomatal guard cells would not match figure 2 if you use Amorphophallous). It will be helpful if the leaf is thick so that a peel can be taken off easily. You could stain the peel using a few drops of diluted red ink or safranin or alta.

Activity-3

Observing cheek cells

Use an ice-cream spoon to gently scrape off a layer(inside your mouth) from the inner walls of your cheek. Do not scrape too hard. Spread the scrapings in a drop of water on a slide. Use a few drops of dilute red or blue ink, safranin or alta to stain the scrapings. Observe the slide under a microscope.

- What are the similarities of cheek cells and cells in the leaf peel?
- Write a difference that you observe between them.

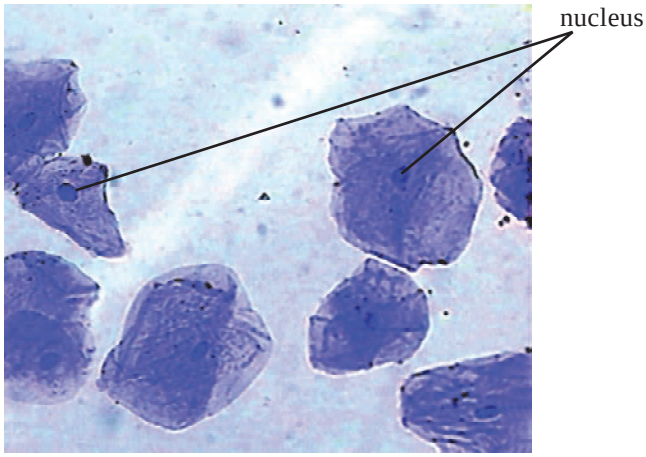


Fig. 3: Cheek cells

We often take the help of a microscope to observe cells as most of them are very small. But there are some large ones like the unfertilized eggs of hen and fish, a juicy hair inside a lobe of citrus fruit like orange etc.Following figure shows us some types of cells of the human body.

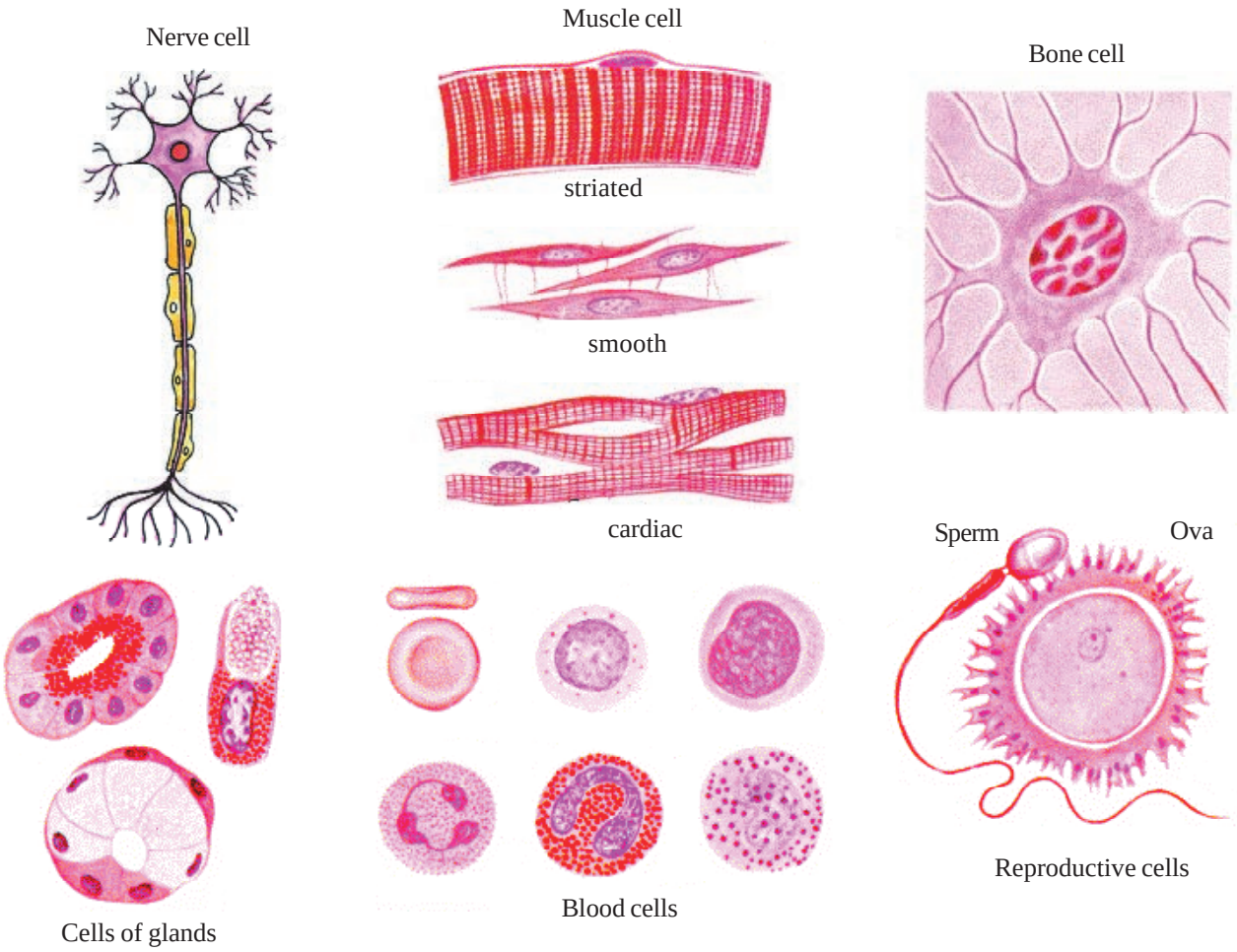


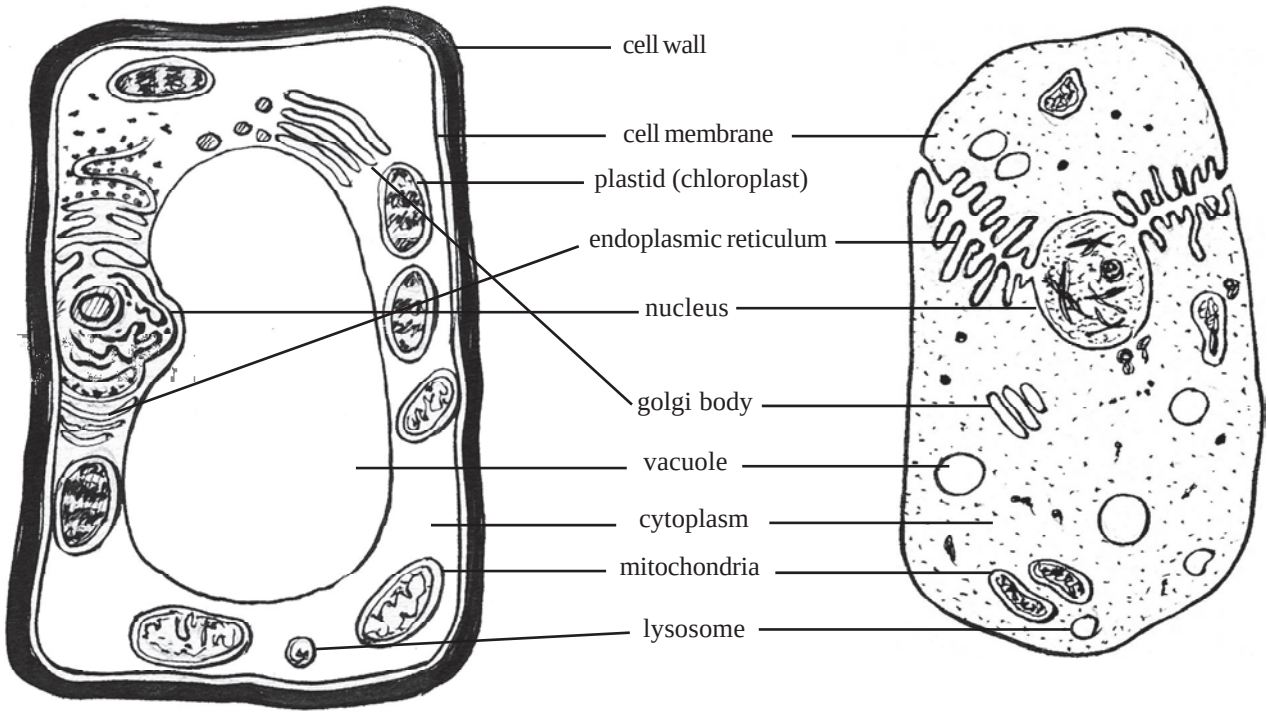
Fig. 4 : Different types of cells

- What differences could you find between these cells?

No matter how different cells appear, be they unicellular organisms or multicellular forms, all life processes (nutrition, respiration, excretion, transport and reproduction) occur in a cell.

6.2 Studying Cells: The Typical Cell

Keeping in view certain similarities of cells, scientists have tried to make models and diagrams to represent their structure. These are called as typical cells. The diagrams of typical animal and plant cells have been given here.



(a) Typical plant cell **(b) Typical animal cell**

Fig. 5

- What are the similarities that you see among the cells?
- Find out a structure present in plant cells but absent in animal cells.

The diagram of a typical cell is drawn on the basis of information gathered from various sources. Usuatly the structure of it is drawn as per the existing knowledge of cells in such a manner that we may get an idea of its appearance. There are several scientists today who treat unicellular organisms as models or typical cells to study cell structure and function. This helps them to get a live and dynamic resemblance to the form of cells. Such a diagram or a model becomes necessary to study cells and their internal structure whether they be our own body cells or those of any other organism.

Do you know?

Within just a period of 200 years we have been able to make microscopes that can give us an enlarged view of cells ranging from 10 to 5 lakh times their actual size. The technology of staining has also improved to such an extent that we are getting deeper insights about the functioning of cells. The structure of cells and their functioning is very complex and our knowledge regarding it is still incomplete, thus research in this area is an ongoing process.

It is not essential that all the parts of cells shown in a typical animal or plant cell may be present at a time in all actual cells. We often see structural variations in cells depending on the function that they carry out. The numbers of cellular parts are also not shown in a typical cell as for example plant cells may have hundreds of chloroplasts, but in a typical plant cell we see only one or two of them. Moreover all plant cells do not have chloroplasts but we see that the diagram of a typical plant cell always contains atleast one. Some parts are of course present in all living cells, but not all parts as shown in a typical cell may be present in all cells. Apart from this, the diagram of a typical cell is not drawn to scale, that is the actual proportion of different parts of the cell are not represented. If we make an attempt to draw a scale diagram of a true cell, the proportion of the cell to its parts could be somewhat like this - if the diameter of the cell is 30 meters, that of its nucleus could be 6meters, a mitochondria would be a meter across and a ribosome merely a centimeter across. This proportion varies from cell to cell. Thus the diagram of a typical cell is only a representative one showing a virtual arrangement.

6.3 Organization of the Cell

The cell has an internal structure that keeps changing. Regarding its parts we have so far come to know that some of them are formed from its membrane itself. We shall study such parts under endomembrane system. There are some other parts that were formed due to the entry of an organism into a cell. The symbiotic coexistence of this part with the cell made it an important part of the cell. We shall study these under endosymbiosis.

- Think and write about the relations that parts of cells have among themselves.

6.4 Endomembrane System and Cell Parts

Endoplasmic reticulum, golgi body, lysosome, cell membrane, nucleus etc. fall under this category. These parts are membrane bound, enclose a space within them that is partioned off from the cytoplasm and usually contain a fluid that differs from it. These parts have contacts and relations among themselves in various ways. One such pathway showing an arrangement of flow of substances is given in figure 6. The direction of arrows represents the flow of substances which can be in both directions.

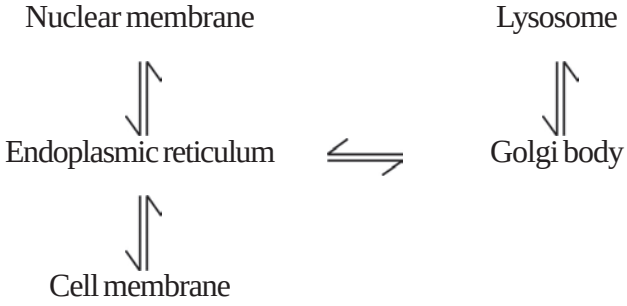


Fig. 6 : The interrelationship of some of the parts of endomembrane system

- Write any three routes though which substances pass to the cell membrane.

Let us now try to know about the structures of some parts of this system in detail.

6.4.1 Cell Membrane

Activity-4

Observation of the cell membrane

Make a concentrated sugar solution(say by dissolving two teaspoonful of sugar in half a cup of water). Put two drops of this on a freshly prepared slide of the peel of Rheo leaf(refer to activity 2). Leave the slide aside for some time (minimum 5 minutes). Now observe it under the microscope.

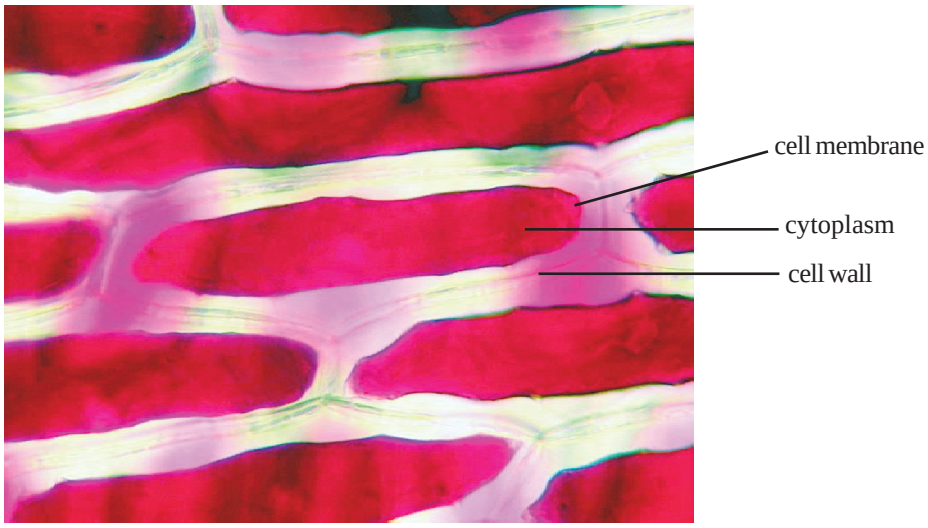


Fig. 7 : Shrinking of cytoplasm in the cell

Did the pink color shrink to a corner of the cell? The outer edge of this colored portion is the cell membrane (refer to diagram of typical cell). The cell membrane along with cytoplasm has moved away from the cell wall. This occurs when the contents shrink and the process is called as plasmolysis.

- How do you think the cytoplasm may have shrunk?

One reason for this is movement of water out of the cell. Does it have something to do with concentration of fluids within and outside the cell?

Suppose there was more water in the sugar solution than that inside the cytoplasm of the cell, water would have moved into the cell. The reverse would occur if concentrations were changed. Thus, the movement of fluids across a membrane due to a difference in concentration is called as osmosis.

Something like this occurs even across the cell membrane which plays a very important role in the passage of materials across it. The structure of the cell membrane is visible only under the electron micro-

scope. It is elastic in nature and made up of mainly proteins and fatty, oily substances called as lipids. It is a bilayer membrane that forms a protective covering of the cytoplasm giving a shape to the cell. Substances are selectively passed across the cell membrane and thus it is called as selectively permeable membrane.

- What would be the effect on a cell if the membrane was not selectively permeable?

Do you know?

There are certain sites of identification on the cell membrane that help cells to recognize each other. This has a lot of importance in the life of organisms. During the early developmental stages, as cells of different layers are formed and displaced, cell to cell connection is established with the help of these sites. Cells recognize each other and connect to play an important role in the formation of tissues and organs. Apart from this, the cells of the body of an organism are able to identify any foreign cells with the help of these sites.

We had tried to observe the cell wall and cell membrane separately with the help of plasmolysis in plant cells in activity-4. We shall now use an unfertilized egg(the common poultry egg) to observe the cell membrane of an animal cell.

Activity- 5

Observing cell membrane in an egg

We would need, an egg, vinegar(circa), beaker, concentrated sugar solution and water for this activity.

Keep the eggs in vinegar for 4-5 days, turn it regularly. The calcium carbonate present in the hard shell would dissolve. A thin translucent covering would be visible and you could touch to feel its texture. This smooth elastic covering is the cell membrane. Now put it in the beaker containing concentrated sugar solution and leave it for around 10 minutes.

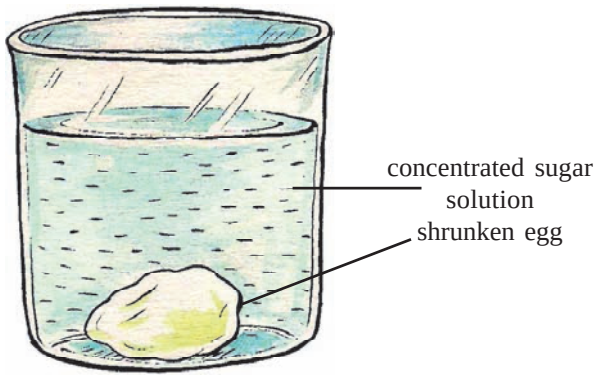


Fig. 8 : A deshelled egg kept in concentrated sugar solution

- What difference do you see in the egg now? What could be the reason for this?

6.4.2 Cell Wall

Cell wall is a special feature of plant cells. Animal cells usually lack a covering outside the cell membrane. Nearly all plant cells have a very prominent mainly cellulosic cell wall that forms a strong covering over the cell membrane. This is a prominent distinguishing feature of plant cells.

It is a hard or elastic and porous covering that renders a particular shape, support and protection to the plant cell. This layer is formed from the live cell. The cells that Robert Hook saw in a section of cork or those of the matchstick are dead cells in which we can see the cell wall.

- What is the difference between cell wall and cell membrane?
- What do you think is the role of cell wall in a plant cell?

6.4.3 Cytoplasm

Cell membrane delimits the extent of the cytoplasm, forming a boundary of it. All other parts of the cell like mitochondria, plastid, endoplasmic reticulum, nucleus etc. are found in the cytoplasm. Thus the cytoplasm is a basic fluid in which, stored, secretory, excretory substances are present. The water content of cytoplasm is usually large.

6.4.4 Nucleus

Robert Brown saw a structure in the center of the cell which he named nucleus in the year 1831. This is a very important part of the cell.

- Observe figures 5 a and b and state whether the nucleus is always located in the center of the cell.
- Which type of cell do you think Robert Brown may have studied, plant or animal?

Do you know?

Some cells do not have a nucleus like the red blood cells in mammals or a type of cell of xylem or phloem tissue(you will read about them in the next chapter) in plants. Actually these cells do contain a nucleus in the initial stages, that degenerates and there is more space available for the transport of substances.

There are thread like structures inside the nucleus which are called as chromatin. They are mainly composed of genetic material. These appear as rods during cell division and are called as chromosomes. Characters from parents are carried to young ones by these structures.

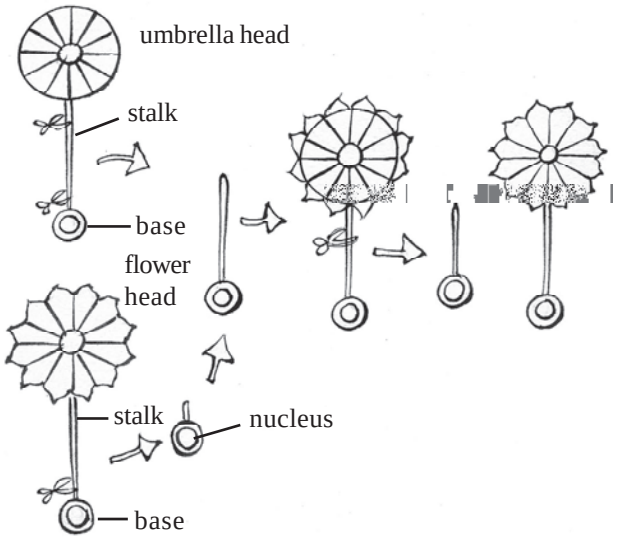
Nucleus controls and coordinates all functions of the cell. It also determines the characteristic features of living organisms. We find a bilayered nuclear membrane around the nucleus that is similar to the cell membrane, in most cells.

Nucleus forms a major basis of classification. Organisms having a membrane bound nucleus are called eukaryotes while those without the membrane are prokaryotes.

Do you know?

Experiment on Acetabularia

A German scientist named Joachim Hamerling experimented on Acetabularia and showed for the first time in 1934 that the nucleus determined all the characters of an organism.



This marine alga is unicellular and has a body with three clearly distinguishable parts, an umbrella like head, a stalk and a base. Different species of the alga have different types of head, like flower shaped, in the form of a reverse umbrella etc. If the head is cut out, it grows again on the stalk.

Hamerling took the base of the alga with a flower shaped head and transplanted the stalk of the algae with an umbrella shaped head. The head of this alga had combined features of both the flower headed and umbrella headed algae. Thus some substances from the base and the stalk determined the form of head. He cut the head of this newly formed alga. The head that formed now was in the shape of a flower. This showed that certainly some substances passed on from the nucleus that determined the shape of the head.

Hamerling had already observed that the base of alga had the nucleus. His experiments along with others on nucleus and cell formation confirmed that the nucleus had the information to form a cell of the body of an organism.

- Can the nucleus be called a control center of the cell? Why?

6.4.5 Endoplasmic reticulum

You will find an extensive network of tubes and vesicles(small rounded bag like structures) in figure 5 a and b. When observed under the electron microscope it appears rough at some places and smooth at others. The figure also shows rough and smooth endoplasmic reticulum. Rough endoplasmic reticulum has ribosomes on its surface and helps in the synthesis of proteins. Smooth endoplasmic reticulum helps in the synthesis of fats. Substances formed with the help of endoplasmic reticulum are constantly involved in the process of membrane repair and synthesis.

- Through which cellular part, can materials be transported from the nuclear membrane to the cell membrane?(take the help of figure 6)

6.4.6 Golgi body

These were first described in the year 1898 by Camillo Golgi and are the only parts in a cell named after a scientist. These are membranous bag like structures There are some fluid filled vesicles along its edges and around it. The proteins synthesized by ribosomes are sent to the golgi body for packaging in vesicles before transport to other sites inside the cell, or outside it.The number of golgi bodies vary from cell to cell. Secretary cells often have large numbers of them.

6.4.7 Lysosomes

There are some lethal substances present in the cells that have the ability to digest the entire cell. How is it that a cell is not affected by them normally? This could be answered with the discovery of lysosomes which were found to be membrane bound bodies that contained the lethal substances. Thus in a normal condition these substances were not in contact with other substances of the cell. The lysosomes destroy certain substances that are produced in the cell or may gain entry into them that are harmful (they are also capable of breaking down larger molecules to smaller ones whether they are harmful or not). Lysosomes themselves may be destroyed in this process. They are also capable of digesting the whole cell, thus they are called as suicidel bags of the cell.

- What do you think will be the role of a lysosome in a completely infected cell?

6.4.8 Vacuoles

- Which is the largest part in the plant cell as seen in Figure 5a?

Fluid filled bodies of various shapes that usually contain storage products are called as vacuoles (term vesicles used earlier represent very small vacuoles). Usually animal cells have smaller vacuoles as compared to plant cells.

6.5 Function of the endomembrane system

Let us take an example to understand the function of this system. This will give us a further idea of how parts within a cell work together to carry out a certain function.

Ribosomes attach to the walls of the endoplasmic reticulum during the process of protein synthesis. The synthesized protein molecule enters the cavity of the endoplasmic reticulum. Small vesicles form from the endoplasmic reticulum that carries these protein molecules to the golgi body. Here packaging of these molecules are done in such a manner that more proteins may be transported together and they may not be damaged during transport. These are then transported to other parts of the cell like nuclear membrane, cell membrane etc. These are used for different purposes like repair or construction of different parts or may be secreted out of the cell for other purposes.

6.6 Parts formed by endosymbiosis

Scientists have found that some parts of the cell were formed due to the entry of a unicellular organism into another. The cell that entered was bilayered membrane bound and had genetic material in it. As time passed these became symbiotic inhabitants in the host cell. They got shelter in the host cell, while the host cell got some important substances like nutrients and energy from the guest cell. These guest cells formed mitochondria or the chloroplast, The former mainly helped in providing energy to the host cell, while the later mainly provided some essential nutrients to it.

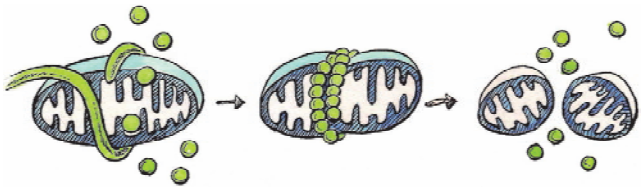
6.6.1 Mitochondria

Different cells have different shapes of mitochondria in them like spherical, rod shaped or branched. These are nearly 6 to 10 times shorter than the nucleus. There can be around 100 to 1000 mitochondria in a cell. 15-20% of a mammalian liver cell consists of mitochondria.

Mitochondria play an important role in the respiration of the cell. It functions as an energy source for running life processes within the cell. Mitochondrial genetic material passes from one generation to the other as mitochondria divide within the cell, during cell division or otherwise.

Do you know?

There is a dynamic relationship of endoplasmic reticulum and the mitochondria. It has been observed that tubules of the endoplasmic reticulum form a noose around the mitochondria stimulating its division. The noose tightens as the mitochondria divide into two parts each of which start functioning as an individual mitochondrion.



- Why is it necessary for mitochondria to be present in large numbers inside a cell?

6.6.2 Plastid

These are mainly found in plant cells. We do not find them in animal cells (follow Whittaker's classification for definition of animals)

Plastids may be of two types on the basis of presence or absence of pigments.

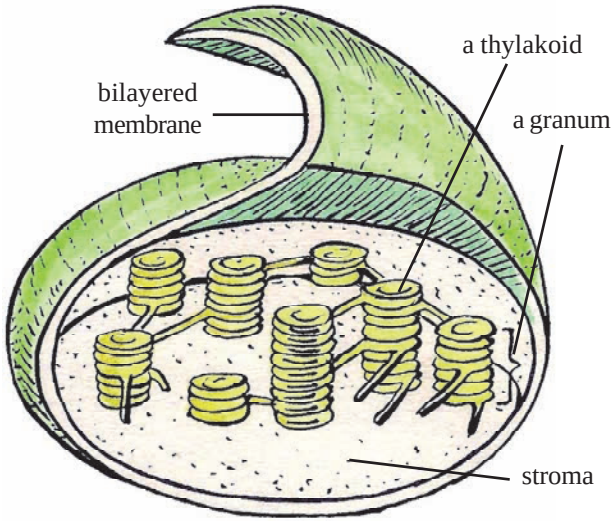
They are-

1. **Colorless plastids-** these lack pigments and are often found as starch filled bodies in maize, potato and radish cells.

2. **Colored plastids-** These contain various pigments and are found in some cells seeds, flowers, leaves, fruits etc. of the plant. The green colored plastid is called as chloroplast. It may be of various shapes like disc, egg, ladder like, ribbon like, star shaped, ring shaped, reticulate etc.

The diameter of a chloroplast is nearly twice that of a mitochondrion. It is bounded by a bilayered membrane. Apart from this there is another membrane that forms coin shaped sac like structures called as thylakoids and their interconnections. Thylakoids arranged in a stack like manner are called as grana.

Stroma is the substance filled between the membranes.



Apparent 3D view of a chloroplast

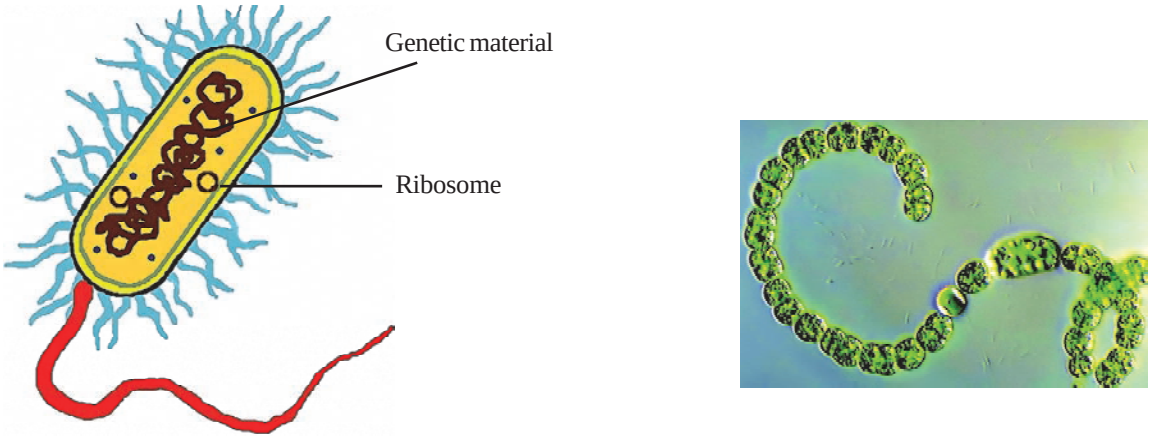
Photosynthesis occurs in plants cells due to the presence of chloroplasts.

- Think why? The chloroplast is often called as the kitchen of the plant cell.
- How would the working of a cell be affected if mitochondria were absent?
- What were the special features of organisms that entered and started living symbiotically in a host cell?

So far we had been studying eukaryotic cells, that is, those containing a membrane bound nucleus. Now let us study prokaryotic cells and their characteristic features.

6.7 Characteristics of Prokaryotic Cells

These cells are bound by a cell wall and a cell membrane. They lack some parts observed in the eukaryotic cell like mitochondrion, endoplasmic reticulum, golgi body etc. The functions of these parts are carried out in the cytoplasm itself. Some of the life processes of these organisms are carried out on the inner side of the cell membrane. These have ribosomes of a smaller size as compared to eukaryotic cells. Most bacteria and cyanobacteria are prokaryotic organisms.



(a) Bacteria (b) Cyanobacteria

Fig. 9: Prokaryotic organisms

- With the help of chapter 1 write other characteristic features of prokaryotic organisms.
- Write at least two differences between eukaryotic and prokaryotic cells.

All life processes like respiration, nutrition, excretion of waste products etc. go on in the cells. All the parts of the cell function in coordination with each other to carry out these processes properly. It is just as all parts of our body function in coordination to carry out these processes in our body. Thus cells have been called the functional unit of living forms.

6.8 Cell Theory

People believed for a long time that life originated spontaneously from air, water and soil. It is only a couple of centuries back, after long periods of research, that we came to know that this was not so. Only living organisms gave rise to other living organisms. But what was that unit from which life forms and the structure of all organisms are made of? Nearly 200 years back research in search of this unit started. Some scientists who largely contributed towards this were Mathais Jacob Schleiden, Theodore Schwann and Rudolf Virchow.

Theodore Schwann was a zoologist while Schleiden was a botanist. Both had extensively studied cells and found that cells were those units that formed the bodies of living organisms. Nearly two decades after this, in the year 1855, Rudolf Virchow found that cells divided giving rise to new cells.

These scientists together proposed the cell theory that became a fundamental theory of life science. The cell theory states that-

1. All living organisms are made up of cells, their communities and intercellular substances.
2. Cells are the structural and functional unit of living organisms.
3. Cells arise from pre-existing cells.

It is important to note that it took nearly 200 years from naming the cell to proposing the cell theory.

- How does cell theory establish our understanding of cells?
- Can we attribute any theory of science to an individual scientist only? Why, why not?

6.9 Why are cells usually small?

Cells are usually so small that it is not possible to see them without a magnifier.

- Have you ever thought why cells may be so small?
- What would happen if cells were big?
- What are the advantages for a cell to have a small size?

We have studied that there is a continuous exchange of materials like nutrients, gases and excretory products between the cell and its surrounding. This happens through the surface of the cells. If the exchange occurs across smaller distances it would require less energy. It could only be possible if the size of the cells were small.

Let us try to understand this with the help of some mathematical calculations -

If we consider a cell to be spherical then, as its radius doubles, the increase in surface area would be in squares and that of volume in cubes.

Let us see the values given in the table. Surface area and volume have been calculated in each case and presented in the table.

Table 1 : The ratio of the surface area to volume of a spherical cell

Radius(mm)	0.5	1.0	1.5	3.0	3.5
Surface Area(SA)	3.4	12.6	28.6	113.4	153.6
Volume(V)	0.6	4.7	14.4	112.5	179.5
Ratio of SA to V	6:1	3:1	2:1	1:1	0.8:1

We may find such relations in different shapes of cells as well.

Observe the table and answer the following questions.

- As the radius of the cell increases what difference do you observe in the surface area and volume of it?
- How does the radius affect the ratio of surface area and volume of cell?

Larger surface area would increase the area of contact with resources outside a cell and thereby also the chance of entry of more of the resources into the cell.

An increase in volume of the cell would increase demand of more resources. We have seen that the increase in surface area is less as compared to the volume. Thus as cell size would increase the demand for intake of resources from the surface would increase.A lesser surface area would fail to meet up to the demands. Thus the pace of life processes would decrease.

You have observed some large cells like that of the eggs of hen (the area with the yolk). These have attained their maturity and their life processes have become very slow. You may have now understood how in spite of their large size they remain alive.

6.10 Are Cells Flat?

Cells appear flat or two dimensional when we observe them under the microscope. No matter how small they might be, cells are actually three dimensional and have length, breadth and height. We may observe the egg cell of birds, or a juicy hair cell of an orange to get an idea of this.

Key words

prokaryote, eukaryote, cell theory, nucleus, mitochondria, chloroplast, lysosome



What we have learnt

- Cells are the structural and functional unit of living organisms.
- Cells were discovered by Robert Hooke.
- Cells are bounded by an elastic, selectively permeable membrane.
- Plant cells have a cell wall, around the cell membrane, made of cellulose.
- Eukaryotic cells have membrane bound nucleus.
- Lysosomes are membrane bound organelles that contain digestive enzymes capable of digesting several types of substances.
- Packaging of substances are done in the golgi bodies
- Mitochondria are the power house of the cell.
- Plastids are found only in plant cells. They are of mainly two groups- colored and colorless.
- Chloroplasts are chlorophyll containing green colored plastids.
- Prokaryotes do not contain membrane bound organelles.
- Cells are not flat they are three dimensional.
- Genetic material is found in the nucleus, mitochondria and plastids of cells.

Exercise

1. Choose the correct option
   - i) Name 'cell' was given by one of the following-
     (a) Robert Hooke (b) Robert Brown
     (c) Leeuwenhoek (d) Flemming
   - ii) Cell theory was proposed by-
     (a) Shleiden, Schwann and Virchow
     (b) Watson and Crick
     (c) Darwin and Wallace
     (d) Mendel and Morgan
   - iii) Single layered membrane is found in-
     (a) Mitochondria (b) Chloroplast
     (c) Lysosome (d) None of the above
   - iv) Cellulosic cell wall is not present in one of the following-
     (a) Bacteria (b) Cells of Hydrilla
     (c) Cells of Mango plant (d) Cells of Cactus
   - v) Prokaryotic cell has one of these-
     (a) Mitochondria (b) Ribosome
     (c) Plastid (d) Lysosome
2. Draw a typical plant cell and label the following parts-
   (a) Cell wall (b) Nucleus (c) Chloroplast (d) Vacuole
3. Write a difference between cell wall and cell membrane.
4. Draw a labeled diagram of a typical animal cell.
5. How are prokaryotic cells different from eukaryotic cells?
6. Explain how the cell theory clarifies our understanding of the cell?
7. Write two similarities and differences between mitochondria and chloroplast.
8. Write the names of those plant parts that contain colorless plastids, chloroplasts, other colored plastids (those other than chloroplast).
9. Explain the function of endomembrane system.
10. Why are cells usually small?
11. We add salt while cooking vegetables. As soon as we add salt we find water coming out. Explain why this happens.
12. What will happen if -
    - (i) There is no nucleus in a cell
    - (ii) Plasma membrane of the cell bursts
    - (iii) Boil a Rheo leaf in water and put it in concentrated sugar solution
    - (iv) Golgi body is removed from the cell

यातायात के प्रमुख कारक

Enforcement (कार्यवाही)
Education (शिक्षा)
Engineearing (निर्माण)
Emergency (आपातकालीन)

Engineearing (निर्माण)

आवागमन को सुगम व सुरक्षित करने के लिए यह आवश्यक है कि आवागमन हेतु सुविधा युक्त सड़क हो। सड़क नाम निर्माण यातायात के अनुरूप होना चाहिए, सड़क निर्माण के दौरान चौंक-चौराहा, लेफ्ट टर्न ठीक हो, मार्ग विभाजक, वाहनों के लिए पार्किंग स्थल आदि यातायात के अनुरूप बना हो तभी आवागमन सुगम व सुरक्षित हो सकता है।

सड़क निर्माण एजेंसियॉं :

(1) राष्ट्रीय राजमार्ग निर्माण विभाग
(2) राष्ट्रीय राजमार्ग विभाग प्राधिकरण
(3) लोक निर्माण विभाग
(4) नगर निगम/नगर पालिक निगम
(5) स्थिानीय विभाग प्राधिकार (ग्रामीण व शहरी)



NATIONAL HIGHWAY

सड़क के प्रकार :

(1) राष्ट्रीय राजमार्ग -**NATIONAL HIGHWAY**
(2) प्रांतीय राजमार्ग -**STATE HIGHWAY**
(3) जिला सड़क- **DISTRICT HIGHWAY**
(4) अन्य सड़क- **OTHER ROAD**



STATE HIGHWAY